THE CURIOSITY SERIES · ARTICLE X

Believe It or Not

The Journey of Water

Science, tradition and human experience — nine dots, one connected picture.

Water is at once chemistry and culture: it carries dissolved minerals, shapes landscapes, sustains the body, and occupies a sacred place in human life. Across traditions, people have asked whether words, prayer, music or intention can change water in ways we do not yet understand. This article does not begin by accepting the claim — or dismissing the people who find meaning in it. It follows nine dots, from the molecule to the mantra, and asks at each one: what can be measured, what can be experienced, and what still needs convincing evidence?

THE QUESTION THAT CONNECTS EVERY DOT

When we say water "responds" to intention, are we describing a change in the water, a change in the person, a meaning — or all three at once?



Dots 1–3 · the molecule, the crystal, the sacred

1

The simplest substance, a profound mystery

Water sustains life, shapes our planet and behaves in remarkable ways at the molecular level. Its molecules form a constantly changing network of hydrogen bonds. Ultrafast experiments have watched that network respond within femtoseconds and settle within about a picosecond [1]; another *Nature* study found it loses its "memory" of its own structure within about fifty femtoseconds [2]. During a two-minute chant, the arrangement inside a glass of water reorganises itself around 10^{14} times. That is genuine complexity — but complexity is not a blank cheque. On its own, it does not show that water understands language or records emotion.

→ If water's structure never stands still, what were the famous crystal photographs showing?

~50 fs
water's structural "memory"

2,000 · 100
senders · blinded judges (2006)

< 0.1%
of snow crystals perfectly symmetrical

2

When water appears to respond to human intention

Masaru Emoto's crystal photographs, Yasuyuki Nemoto's work on *hado* and the documentary *Every Living Thing* popularised a striking idea: water exposed to love, gratitude or music forms beautiful, symmetrical crystals; anger or harsh sound, fragmented ones. The evidence deserves a fair reading. In a 2006 double-blind pilot, about 2,000 people in Tokyo directed intention at water in a shielded room in California, and 100 blinded judges rated the treated crystals as more beautiful [3]; a 2008 triple-blind replication also reported a difference [4]. But what was measured was how appealing a photograph looked — not a property of the water. Crystal shape is already governed by temperature and humidity: a fraction of a degree moves a crystal from plate to column [5].

→ Science measured a photograph. Tradition never claimed a photograph — it claimed meaning.

3

The ancient Indian connection

In Hindu traditions, water is more than a substance. The *kamandalu* carried by sages, the *kalasha* of worship and the *abhisheka* of blessing make water a medium of purification, intention and devotion; stories of *tapas* and *sankalpa* connect intention with sacred action. That meaning is real as human tradition. A claim that water physically stores a blessing is a different kind of claim — about mechanism — judged by different evidence.

→ So where does tradition meet something we can measure? In a copper vessel.

THE DISTINCTION THAT MATTERS

A positive experimental result can be evidence for a specific effect under a specific method. It does not automatically prove the interpretation attached to that effect.



4

Copper: ancient practice, measurable science

In a 2012 study, water carrying six diarrhoeagenic bacteria and stored 16 hours in copper pots yielded no recoverable bacteria; dissolved copper measured 177 µg/L [6]. The mechanism is named: copper ions damage membranes, drive reactive-oxygen chemistry and disrupt iron–sulphur machinery [7]. Limits: it depends on water quality and contact time; WHO's guideline is 2 mg/L [8]; never store lemon water or other acidic drinks in copper; it is no substitute for clean, treated water — and it says nothing about stored prayers.

→ Copper changes water chemically. Can sound change it too?

5

Mantras, sound waves and vibration

A chanted mantra is also physical sound — pressure waves with a measurable frequency that can make nearby water vibrate. Vibrating-plate patterns are real physics: standing waves set by frequency and vessel shape, not by the meaning of the words. In physics, "vibration" is measurable motion; in spiritual language, presence and devotion. Both are meaningful; they are not interchangeable. There is no scientific basis for calling the Gayatri Mantra the "longest recorded frequency in the world."

→ The sound reaches the water. Its meaning reaches the person.

6

Hanuman Chalisa: water as devotion

Water placed before Hanuman Ji during the Hanuman Chalisa, then taken as *prasad*, expresses faith, protection, courage, surrender and hope — and can bring real calm and focus. The act is observable: a text is recited, a vessel is present, water is consumed. A new physical property in the water is not automatically observable; it would have to be measured independently and shown to be reproducible.

→ Which leads to the oldest question of all: can water remember?



7

Can water remember?

"Memory of water" was born in 1988, when *Nature* published Jacques Benveniste's claim that an extremely diluted solution still triggered a biological response — with an editorial reservation attached [9]. Rerun with the codes sealed in foil and taped to the ceiling, the effect did not appear [10]; later tests did not reproduce it consistently [11]. Past about the twelfth hundredfold dilution no original molecule remains — hence the word "memory." Water does carry a memory of its composition: minerals, gases, isotopes, the vessel it touched. A memory of information — a word, an emotion — remains unproven.

→ If the water may not remember, why do people feel better?

8

Belief, expectation and healing

Several explanations can coexist: reduced stress, slower breathing, positive expectation, changed behaviour, natural recovery — and, for many, grace. A 2025 review of 24 neuroimaging studies found chanting engages brain regions for attention and emotional regulation [12]. Expectation is itself a mechanism: placebo pain relief can be blocked by naloxone, which blocks the brain's own opioids [13], and patients told openly they were taking sugar pills still reported relief [14]. A meaningful experience is real — but on its own it does not prove the water heals. Devotional water can complement medical care, never replace it.

→ Now the dots can be connected.

9

Connecting the dots

Water is biological necessity, chemical medium, cultural symbol and spiritual offering. Copper is a tradition with a measurable mechanism. Chanting has measurable effects on the person. Sacred-water traditions carry meaning that science does not need to validate for it to be real. And some questions — does water retain intention? can a mantra leave a lasting trace? — remain open. The honest position is neither automatic belief nor reflexive dismissal: keep every claim at the level its evidence supports.

ESTABLISHED

- Water's structure rearranges in femto- to picoseconds
- Sound moves liquids
- Copper inhibits microbes under controlled conditions

CONTEXTUAL

- Chanting shifts attention and stress
- Expectation changes the body
- Tradition gives water meaning

STILL OPEN

- Water storing intention
- Prayer-charged water healing by itself
- A lasting "memory" of words

BELIEVE IT OR NOT

Let every dot remain a dot until the evidence connects it. Curiosity is not the promise that every mystery will confirm our beliefs; it is the willingness to investigate without deciding the answer in advance.

This article is an exploratory reflection, not medical advice. Sacred water and devotional practices should not replace potable water, medical diagnosis or evidence-based treatment.

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